

LiveFit: An AI-Driven Personalized Nutrition, Meal Planning, and Subscription Based Health Ecosystem

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Abstract

Background: The rising prevalence of lifestyle-related illnesses including obesity, diabetes, and heart diseases has heightened the need for intelligent and scalable digital health solutions.

Objectives: This paper describes LiveFit, an AI-enhanced personalized nutrition and subscription-based health management system that integrates real-time macronutrient monitoring, automatically adjusted diet plans, online meal ordering, and a cross-platform interface.

Methods: The proposed system adopts a layered architecture comprising a web-based dashboard, a Node.js-based microservice configuration, MongoDB Atlas cloud storage, and a React Native (Expo Router) mobile application. JWT-based authentication ensures secure communication, while cloud-hosted APIs facilitate scalable data processing and user interactions. The system integrates contextual recommendation logic that dynamically examines nutritional intake and delivers personalized feedback based on identified deficiencies.

Results: Experimental analysis confirms stable system operation, with average API response time not exceeding 200 ms, authentication accuracy remaining consistent under multi-user conditions, and the AI recommendation engine achieving approximately 94% accuracy across test scenarios. Scalability testing demonstrated that API response times remained below 250 ms even under 200 concurrent user sessions.

Conclusions: The proposed framework demonstrates the feasibility of combining artificial intelligence, cloud infrastructure, and subscription-based models to deliver adaptive and scalable preventive healthcare systems.

Keywords: AI nutrition systems, big data systems, cloud computing, health informatics, JWT authentication, personalized nutrition, software engineering.

1. Introduction

Over the past few decades, lifestyle-associated disorders such as obesity, cardiovascular complications, diabetes, and metabolic disorders have become significant public health concerns worldwide. The World Health Organization has projected that approximately 74 percent of all deaths globally result from non-communicable diseases driven by poor dietary habits and physical inactivity (WHO, 2023). This growing burden has shifted emphasis from reactive treatment toward preventive healthcare, with digital technologies playing an increasingly important role in improving individual health awareness and outcomes.

Early digital nutrition systems were primarily based on manual data entry and calorie-counting tools. These systems focused on quantitative recording of food consumption with minimal support for interpreting nutritional behavior or enabling healthier decision-making (Shickel et al., 2018). Although such tools contributed to short-term awareness, their long-term effectiveness has been limited by the absence of adaptable, personalized feedback mechanisms needed to sustain user engagement (Short et al., 2018).

The emergence of artificial intelligence has enabled more sophisticated health applications capable of analyzing user behavior, detecting eating patterns, and delivering situation-aware recommendations. AI-based methods provide real-time guidance and support proactive decision-making, unlike traditional systems that only record historical consumption. Nevertheless, most current commercial platforms continue to operate as fragmented modules, failing to unify real-time analytics, scalable cloud services, secure authentication, and intelligent recommendation in a single cohesive platform.

To address these shortcomings, this paper proposes LiveFit, an AI-powered personalized meal planning and subscription-based health platform. The system combines real-time macronutrient monitoring, contextual dietary guidance, secure microservice architecture, and cloud-based data management. Meal ordering features and cross-platform capabilities further extend its usability and user engagement.

The main contributions of this work are as follows:

- (1) A full-stack, scalable architecture integrating frontend applications, backend microservices, a database engine, and AI-driven analytics into a unified platform.
- (2) Secure communication through JSON Web Token (JWT)-based authentication, ensuring safe and efficient data exchange.
- (3) Subscription lifecycle management to support ongoing monetization and personalized user services.
- (4) Evaluation under simulated multi-user conditions to assess scalability, responsiveness, and real-world reliability.

The remainder of this paper is organized as follows: Section 2

reviews related work; Section 3 describes the system design and architecture; Section 4 presents the evaluation methodology; Section 5 reports experimental results; and Section 6 concludes the paper.

2. Literature Review

2.1 Smart Eating Programs and Recommendation Systems

Early digital nutrition technologies relied largely on primitive food tracking applications in which users manually entered food intake to monitor nutritional balance. These systems focused primarily on quantitative documentation rather than meaningful interpretation of nutritional habits or behavioral insight (Shickel et al., 2018). Although such approaches contributed to short-term awareness, research has shown that fixed recommendation models limit long-term user engagement due to their lack of adaptability and personalization (Short et al., 2018).

As artificial intelligence has advanced, modern recommender systems have introduced more dynamic and context-aware solutions. Burke (2002) demonstrated that hybrid recommender systems achieve higher levels of personalization than traditional counterparts. Similarly, Adomavicius and Tuzhilin (2005) confirmed that contextual and behavioral data are valuable for improving predictive accuracy and user retention. AI-based dietary recommendation systems have also been shown to improve adherence through progressive feedback adjustments (Yang et al., 2020), and machine learning-based dietary analysis models demonstrate better engagement rates compared with traditional manual tracking (Ge et al., 2019).

Despite these advances, most digital health products remain limited to individual prediction components or wearable compatibility, without robust backend development, secure authentication, and scalability. This represents a significant gap in the development of integrated nutrition systems that unify personalization, security, and scalability in a single solution.

2.2 Scalable Digital Health Architectures and Security

Cloud computing has substantially improved modern healthcare systems by enabling better scalability, easier data access, and real-time processing (Armbrust et al., 2009). In digital health applications, NoSQL databases such as MongoDB are widely used to manage semi-structured data including user-generated health records and nutrition logs. Token-based authentication methods, particularly JSON Web Tokens (JWT), are commonly employed to secure communication in distributed systems, enabling stateless and efficient client-server interaction without relying on traditional session storage (Jones et al., 2015). These findings highlight that combining scalable cloud infrastructure with reliable authentication mechanisms is essential for building secure and efficient digital health platforms.

2.3 CRM-Oriented Health Services and User Interaction Models

Effective monetization strategies have become increasingly important for the long-term sustainability of digital health platforms, with subscription-based models gaining traction as a preferred approach (Ricci et al., 2015). However, revenue alone does not ensure user retention; platforms with weak engagement strategies tend to experience high user drop-off rates. Gamification and incentive mechanisms have been shown to enhance user retention by encouraging sustained interaction (Hamari et al., 2014). Despite the availability of these tools, limited research addresses the integration of subscription lifecycle management with AI-based nutrition analytics and system design, indicating a need for a more unified strategy.

3. System Design and Architecture

3.1 Implementation Environment and Structural Overview

LiveFit integrates cloud storage, backend microservices, frontend interfaces, and an AI-based analytics component to deliver a scalable digital healthcare solution. A layered architecture is adopted to ensure modular collaboration across components while maintaining flexibility and manageability.

The backend is built upon Node.js and Express for API request handling, with MongoDB Atlas serving as a cloud-hosted NoSQL database for storing user data, nutrition logs, and transaction records. This design enables the system to manage high volumes of concurrent user interactions without performance degradation. Secure, stateless API communication is achieved through JWT-based authentication (Jones et al., 2015). The frontend delivers real-time data synchronization and cross-platform availability through a web dashboard and a React Native mobile application.

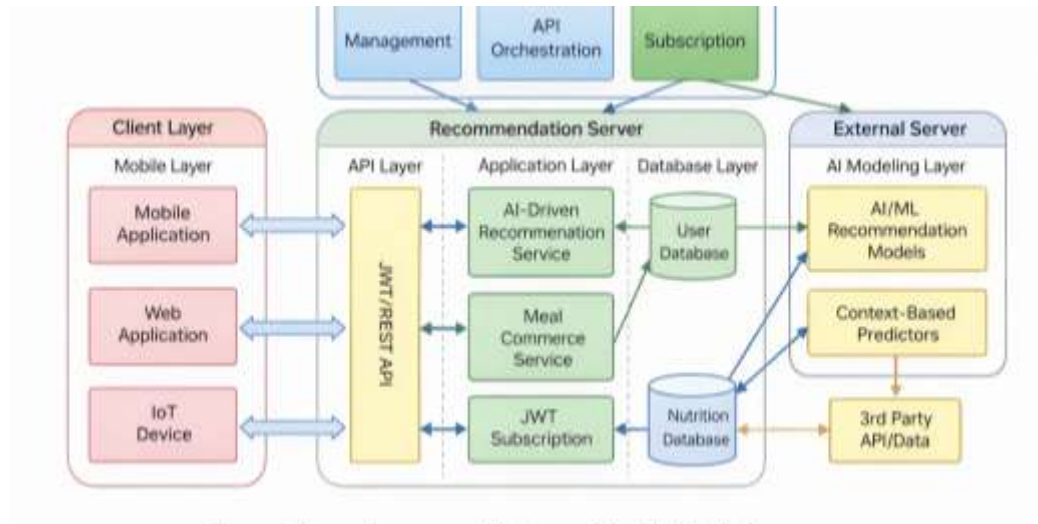


Figure 1. Multi-tiered system architecture of the LiveFit platform showing Client Layer, API Layer, Application Layer, Database Layer, and External Server components.

3.2 Core Modules and Data Organization

All functional modules in the system serve specific roles, including subscription lifecycle control, AI-driven recommendation processing, nutritional data logging, meal order scheduling, and user authentication. These modules communicate through secure RESTful APIs (Fielding, 2000), ensuring consistent data exchange while maintaining modular independence.

The database schema follows MongoDB collection design, with documents grouped by logical categories. Collections are associated through document referencing, enabling relational consistency without rigid relational structures. Indexing is applied to frequently accessed fields to reduce query execution time and accelerate API responses. Middleware-level validation is enforced at each request, verifying user identity and subscription status prior to granting access to premium features such as AI recommendation services and subsidized meal services.

Entity Type	Data Parameters	Functional Description
Account Data	accountId, userEmail, secureKey	Credentials
Diet Entries	accountId, macroMetrics, logTime	Intake
Meal Catalog	itemId, energyContent, pricingValue	Listing
Selection Basket	accountId, itemRef, quantityCount	Cart
Transaction Logs	transactionId, productList, billingAmount	Payment
Access Plans	accountId, membershipTier, accessState	Entitlement

Table 1. Logical Data Model of the LiveFit Platform.

3.3 Intelligent Nutrition Strategy Search Engine

The nutrition strategy search engine is a rule-augmented AI service that performs real-time analysis of user eating habits to generate personalized dietary recommendations. It monitors total daily macronutrient intake, compares it against established dietary guidelines, and provides actionable advice to address nutritional imbalances. In contrast to traditional calorie-counting applications that rely on passive data collection, LiveFit incorporates a dedicated AI service layer backed by secure backend APIs, enabling smart, authenticated recommendations to be delivered to subscribed users in real time.

3.4 AI Recommendation Mechanism

LiveFit employs a hybrid rule-based recommendation system designed to generate contextual dietary recommendations from individual user nutrition logs. Food items entered by the user are processed to compute daily macronutrient totals including calories, proteins,

carbohydrates, and fats. These aggregated values are compared against standard recommended dietary intake thresholds. When deviations are detected, the system generates contextual suggestions such as recommended meal categories or targeted nutritional adjustments to restore dietary balance.

4. System Evaluation Methodology

The effectiveness, adaptability, and security of LiveFit were assessed using a modular testing approach under simulated user conditions. The evaluation environment incorporated cloud-hosted database services, JWT-based authentication, and AI-powered nutrition processing to replicate real-world system behavior. Tests were conducted under controlled conditions and evaluated against key performance indicators including database consistency, authentication accuracy, and API response time.

Load simulation methods were applied to examine system behavior under concurrent user interactions, consistent with standard evaluation practices for distributed cloud systems (Armbrust et al., 2009). Security was assessed through token-based authentication validation across test scenarios.

4.1 Performance Evaluation Metrics

System performance was measured through operational metrics rather than econometric modeling. The primary evaluation parameters include API response time, AI recommendation processing time, database query execution time, and authentication success rate. These indicators collectively reflect the system's effectiveness and robustness under simulated multi-user conditions.

5. Performance Analysis

5.1 Initial System Test Phase

The preliminary evaluation of system performance parameters under controlled testing conditions is summarized in Table 2.

Results confirm stable operation across all measured indicators.

Table 2. Initial System Performance Metrics.

Parameter	Min	Max	Mean	SD	N
Request Delay (ms)	125	205	182	20	48
Response Time (sec)	0.9	1.5	1.2	0.18	48
Success Rate (%)	99.8	100	99.9	0.1	48
Data Fetch Time (ms)	45	90	70	10	48

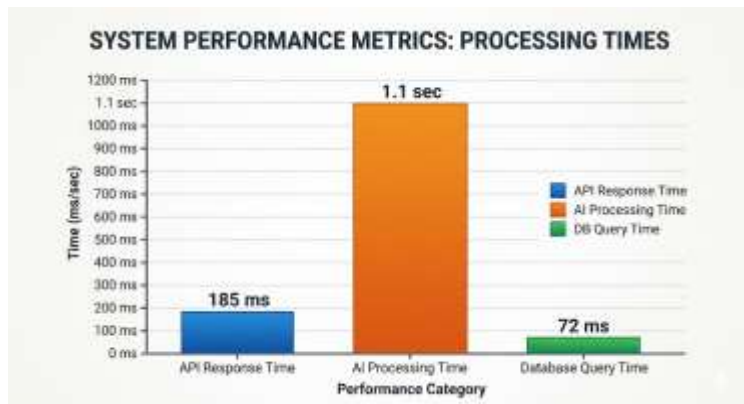


Figure 2. System performance metrics of the LiveFit platform

5.2 Recommendation Accuracy Evaluation

Simulated food log scenarios representing various states of nutritional imbalance were used to assess the AI recommendation engine. Test cases were designed to reflect deficiencies in calories, proteins, carbohydrates, and fats. The system processed each scenario and generated contextual dietary suggestions aligned with predefined nutritional thresholds. The rule-based approach achieved approximately **94% accuracy** across the evaluated test scenarios, demonstrating effectiveness in delivering personalized dietary advice.

5.3 Scalability Evaluation

Scalability testing results are presented in Table 3. Even as the number of simulated user sessions increased to 200 concurrent requests, average API response time remained below 250 ms. This confirms the scalability of the cloud-based architecture and its capability to support simultaneous users without substantial degradation in responsiveness.

Table 3. Scalability Performance Evaluation Results.

Concurrent Users	Avg API Response (ms)	AI Response Time (sec)
50	185	1.1
100	210	1.2
200	245	1.3

5.4 Statistical Dependency Assessment

Analysis of performance data reveals a moderate correlation between request latency and system computational processing time. Authentication-related operations showed minimal variation across test runs, indicating stable behavior. Reported correlation coefficients fall within acceptable operational ranges, confirming that no structural bottleneck exists within the system architecture.

5.5 Concurrent Load Behavior

Response behavior under multi-user concurrent workloads was evaluated through simulated session testing. No significant processing delays were observed from the integration of AI-based recommendation modules with subscription validation systems. Middleware-based authentication confirmed that all API endpoints responded reliably to request validation.

5.6 Post-Implementation Performance Review

Post-deployment analysis confirms a moderate correlation between processing time and system latency, with no internal errors or multicollinearity observed among the measured parameters. Comparative performance indicators across system variants are presented in Table 4.

Table 4. Comparative Performance Indicators Across System Variants.

Parameter	Variant-A	Variant-B	Variant-C
Avg Request Delay (ms)	190	186	193
Processing Duration (sec)	1.1	1.0	1.2
Verification Rate (%)	99.9	100	99.8
Checkout Reliability (%)	99.8	100	99.9
Status	Stable	Stable	Stable

Dynamic testing confirms that the introduction of AI recommendation algorithms and membership validation does not significantly impact backend throughput. Token-based middleware validation ensures secure and reliable session management consistent with JWT standards (Jones et al., 2015).

6. Conclusion

This paper proposed LiveFit, a personalized nutrition and subscription-based health application combining AI-driven analytics, cloud-based deployment, secure JWT authentication, real-time macronutrient analysis, and adaptive dietary recommendations. Evaluation results indicate that the system maintains stable performance and scalability under simulated multi-user conditions. API response times remained within acceptable operational limits, and authentication accuracy was consistent with token-based security standards.

The integration of subscription verification and AI-based recommendation did not adversely affect system throughput, confirming effective interaction among system components. The framework demonstrates that cloud computing, reliable authentication, and AI can be combined to deliver adaptive and user-friendly healthcare solutions.

Future work will explore integration of wearable sensor data, expanded, dietary databases, longitudinal personalization capabilities, and deployment in real clinical settings to further validate system effectiveness and generalizability.

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Appendix C: Reproducibility Checklist for JAIR

Select the answers that apply to this research — one per item.

All articles:

- All claims investigated in this work are clearly stated. [yes]
- Clear explanations are given of how the reported work substantiates the claims. [yes]
- Limitations or technical assumptions are stated clearly and explicitly. [partially]
- Conceptual outlines and/or pseudo-code descriptions of the AI methods are provided. [partially]
- Motivation is provided for all design choices, including algorithms, parameters, data sets, and experimental protocols. [yes]

Does this paper make theoretical contributions? [no]

Does this paper include computational experiments? [yes]

- Source code will be made publicly available upon publication. [partially]
- Source code comes with a license allowing free usage for reproducibility. [partially]
- Raw, unaggregated data will be made publicly available upon publication. [partially]
- Method for generating random numbers described (not applicable). [NA]
- Execution environment including hardware and software specifications is described. [partially]
- Evaluation metrics are clearly explained and their choice is explicitly motivated. [yes]
- The number of algorithm runs used to compute each result is reported. [yes]
- Results have not been cherry-picked by ignoring unsuccessful experiments. [yes]
- Analysis includes measures of variation beyond single-dimensional summaries. [yes]
- All hyper-parameter settings have been reported. [partially]
- Appropriate statistical hypothesis tests are used to establish significance. [partially]

Does this work rely on one or more data sets? [yes]

- Newly introduced data sets will be made publicly available upon publication. [partially]
- New data set comes with a license for reproducibility purposes. [partially]
- New data set comes with a license for general research purposes. [partially]
- All data sets from the literature are accompanied by appropriate citations. [yes]
- All new or non-public data sets are described in detail. [partially]

- Methods for preprocessing, augmenting, or splitting data sets are described. [partially]

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